

HS 312 – Introduction to Science and Technology Studies

Lecture 7

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The actual conduct of science

- Are the norms of science constant through history and across science? *Are they distinctive to science?*
- Norms can be interpreted differently by different actors
- Universalism, disinterestedness (e.g. rationalism), and organized scepticism are also professed by societies for many activities – not distinctive to science
- *Social vs cognitive norms*
- Science as governed by a paradigm ; normal science is shaped by solidarities built around key ideas
- Immanuel Velikovsky's book on historical catastrophe was termed pseudo-science; Mulkay saw it as violation of universalism and organised scepticism because the criticism was levelled without reading the content

The actual conduct of science

- What if secrecy, particularism, interestedness, and credulity were common?
- Science research for example is competitive and work is linked to financial stakes
- Ian Mitroff conducted a study of scientist working on Apollo moon project
- **Counter-norms** – scientists approved commitment and called disinterestedness a myth; claims are valued in terms of who makes them; secrecy is valued; dogmatism allows people to build on previous work
- So **analytical framework of norms does no work** if norms and counter-norms exist

Interpretation of norms

- Norms have to be interpreted
- Mulkay observed in Velikovsky case that depending on context norms can be interpreted as violated or adhered to
- Based on previous sensational claims, scientists could be seen as practising organised scepticism
- By creatively selecting contexts, any scientist can use the norms to justify almost any action - norms do not constrain scientists

Norms as resources

- If norms are interpreted flexibly then we must study how they are used
- Einstein's refusal to accept the truth of quantum mechanics, is often seen as being conservative and violation of disinterestedness
- That is to say Einstein was wrong and QM was right – but can we ignore critics of QM?
- Norms are used in this case to eliminate conflicting views (without attention to argument)
- Opponents of QM are apt to see Einstein maintaining a youthful scepticism
- STS hence sees norms as rhetorical resource; shift from structure-centred perspectives to more agent- or action-centred perspectives

Boundary work

- Issue of epistemic authority
- Boundary mandates that some people do not have the authority
- Source of authority for science – rationality, connection to nature/technology/policy
- But it is a result of an effective boundary work – successful effort to define science in terms of rationality
- People, methods, qualifications are used for charting boundaries

The actual conduct of science

2	22B1512	Adhiraj Joshi	Chapter 3		12	22B1833	Namala Aadarsh
	22B2215	Aditya Meena	The sun in a test tube: the story of cold fusion	29-Jan		23B1852	Neha Kanwadiya
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Summary

- Unless flexibly interpreted, norms do not describe the behaviour of scientists
- But the flexibility means that norms do not do explanatory or analytic work
- The idea that science is a unified institution with an overarching goal is contested
- Could an overarching goal for science have any effect on the actions of individual scientists?
- Critics suggest that science is better understood as the combined product of scientists acting to pursue their own goals
- Merton's norms are ideological resources for moral economy of science, and serve as organizational myth

Sources

- Sismondo Sergio (2010). Questioning Functionalism in the Sociology of Science. Chapter 3, in *An Introduction to Science and Technology Studies*; second edition: Blackwell Publishing, Oxford UK.
- Harry Collins and Trevor Pinch (1998). The sun in a test tube: the story of cold fusion. Chapter 3, in *The Golem: what you should know about science*; second edition: Cambridge University Press, Cambridge UK.

Thank you